

Main Document

Text-to-Image Evaluation for India — E-commerce Product Photography

1. What Eval I Choose

I designed (<https://joshtalks-ai-image-evaluation.onrender.com/>) and ran a blind, human-judged comparison of three text-to-image models — OpenAI GPT Image 1, Google Gemini 2.5 Flash Image, and Google Gemini 3.1 Flash Image (Preview) — on a single, focused task: generating catalog-ready product photos for small apparel and handicraft sellers who sell on Meesho, Instagram, and WhatsApp Business.

Participants rated anonymized outputs (labelled A/B/C, randomized per prompt) across four axes — prompt relevance, commercial usability, Indian-context fit, and visual quality — and made one forced-choice pick per prompt: “which would you actually post on your own store?”

2. Category and Use Case

Category: E-commerce product photography.

Use case: Hero and catalog images for solo apparel/handicraft sellers — sarees, jewellery, festive wear, home décor — the single largest social-commerce category in India.

Seven prompts were used, each describing a distinct product with enough detail to be unambiguous, spanning garments, jewellery, flatlay products, and festive/seasonal styling so the eval wasn't overfit to one product type:

- Maroon silk saree — festive catalog shot with gold Zari border
- Oxidized silver jhumkas — close-up jewellery catalog shot
- Mustard embroidered kurta set — Diwali sale banner mood
- Handwoven jute tote bag — flatlay catalog shot
- Silver anklets — close-up WhatsApp catalog shot
- Light blue men's kurta — studio catalog shot
- Brass diya set — festive home décor listing

3. Why I Chose This

Existing public benchmarks (Arena AI, Artificial Analysis Image Arena) already test general image quality and aesthetics well. What they don't test is something much more specific and commercially meaningful: would a real Indian small seller actually post this image on their own store? That's a different, sharper question than “is this a nice picture,” and it's the one most directly tied to whether these models create real economic value for the people who'd use them.

4. Why It Is Useful for India

Crores of small sellers depend entirely on Meesho, Instagram, and WhatsApp Business for sales, most shooting product photos on their own phones with no studio, no lighting setup, and no design budget. A model that can turn a rough phone photo and a text description into a convincing, sellable product shot is a direct, practical unlock for seller income — not a novelty use case.

This use case also surfaces failure modes that matter specifically in the Indian context and that generic global benchmarks tend to miss: incorrect saree drape, unnatural skin tones, generic/Western-coded backgrounds and styling, and garbled text on packaging or signage.

5. Why It Matters for AI Labs Building for India

This is a genuine product wedge, not a demo. Whichever model wins here is the one worth bundling into seller-facing tools — Meesho's seller app, WhatsApp Business catalog tools, or a Google seller ecosystem product — with a direct line to seller revenue and adoption.

It also tests dimensions a general-purpose leaderboard won't: representation and cultural fit (skin tone, drape, festive context), not just raw photorealism or prompt adherence.

6. How the Evaluation Works

6.1 Models and generation process

All three required models were used, with the same prompt text sent to each. Two access issues came up during generation and are worth documenting as findings in their own right, not just logistics:

- GPT Image 1: ChatGPT's consumer app now defaults to a newer image model rather than the exact GPT Image 1 snapshot. This was confirmed by generating the same prompt via LMArena's Direct Chat mode (which allows pinning the exact model gpt-image-1) and comparing — outputs differed meaningfully from the ChatGPT app version. Final images use the LMArena-verified gpt-image-1 output.
- Gemini 2.5 Flash Image: this model has been effectively deprecated from Google's free consumer surfaces (Gemini app, AI Studio's default picker) in Favor of Gemini 3.1 Flash Image (“Nano Banana 2”). A direct API test via a Collab notebook returned a confirmed RESOURCE_EXHAUSTED error with an explicit free-tier limit of 0 requests for this model — i.e., it is paid-only via the API with no free allowance at all. This was resolved by generating this model's images via LMArena instead, which still hosts it for direct comparison.

6.2 Blind randomization

For every prompt, which model appears as Option A, B, or C was independently randomized, so no participant could learn a pattern across prompts. Model identity was never shown to raters at any point during scoring.

6.3 Real-time data flow

Ratings are not batch-processed or manually collected — every submission writes to a live Google Sheet the moment a participant finishes, and the Results dashboard reads directly from that same sheet, so the leaderboard and axis chart are always showing the current state of the data, not a snapshot:

- [Google Sheet as backend data storage.](#)
- A participant completes all 7 prompts in the rating tool and clicks “Submit all ratings.”
- The tool sends the participant's consent details and all 21 ratings (7 prompts $\times$ 3 options) as one request to a Google Apps Script Web App.
- The Apps Script appends one row per rating to a “Ratings” Google Sheet — 21 new rows per submission — timestamped automatically.
- When anyone opens the Results tab, the tool sends a separate request to the same Apps Script, which reads every row currently in the sheet and returns it as data.
- The tool aggregates that data client-side into the leaderboard, the per-axis score breakdown, and the win-rate figures, and renders them immediately — no manual export or recalculation step.

Because the Sheet is the single source of truth, the same data can also be reviewed directly in Google Sheets at any time — useful both as a live audit trail and as the raw file included in supporting material.

7. How Participants Judged the Outputs

Ratings were collected through a purpose-built rating tool (also submitted as the UI/dashboard/leaderboard deliverable). For each of the 7 prompts, participants saw all three anonymized options side by side and:

- Scored each option 1–5 on four axes: prompt relevance, commercial usability, Indian-context fit, and visual quality
- Could zoom into any image (magnifier control, up to 400%) to inspect fine detail such as text rendering or drape accuracy before scoring
- Made one forced pick per prompt: “which would you actually post on your own store?” — the single strongest preference signal
- Could navigate back to a previous prompt without losing earlier answers

8–10 participants took part, each confirming they were 18+ and giving explicit consent — name, email, age, and agreement — before rating anything, using the consent statement provided in the brief. Consent and rating data were collected and exported separately as supporting material.

8. What Results I Found

8.1 A data-quality check worth noting

Before analysing results, I checked the raw export for anomalies and found one participant's full submission (all 21 rows — 7 prompts $\times$ 3 options) had been recorded three times, identically, almost certainly from a

double/triple click on “Submit all ratings” rather than three genuine responses. Left uncorrected, this would have given that one participant's ratings 3× the influence of everyone else's in the leaderboard math. I deduplicated by (participant, prompt, option) before computing any results below — keeping one real participant record.

Final dataset: 10 unique participants × 7 prompts × 3 options = 210 individual ratings, 70 forced picks total.

8.2 Leaderboard

Model	Avg. score (/5)	Win rate	Picks
Gemini 3.1 Flash Image	3.93	39%	27 / 70
GPT Image 1	3.83	27%	19 / 70
Gemini 2.5 Flash Image	3.83	34%	24 / 70

8.3 Average score by axis

Model	Relevance	Usability	India fit	Visual
Gemini 3.1 Flash Image	3.94	3.96	3.99	3.84
GPT Image 1	3.71	3.83	3.91	3.84
Gemini 2.5 Flash Image	3.80	3.79	3.84	3.87

8.4 Findings

- Gemini 3.1 Flash Image was the clear leader on both metrics that matter — highest average score (3.93/5) and highest win rate (39%) — and led on every individual axis except visual quality, where it tied with GPT Image 1.
- GPT Image 1 and Gemini 2.5 Flash Image scored almost identically on average (3.83 vs. 3.83) but diverged noticeably on win rate (27% vs. 34%) — exactly the kind of split the two metrics are meant to catch: Gemini 2.5 Flash Image was more often the one participants would actually choose to post, even though its average rating wasn't higher.
- “Indian context fit” scored highest of all four axes for every model (3.84–3.99), suggesting none of the three models struggled badly with representation or cultural fit on these prompts — a genuinely reassuring finding given this was the axis I was most worried about going in.
- GPT Image 1 scored lowest specifically on prompt relevance (3.71, versus 3.80–3.94 for the Gemini models) — worth a closer qualitative look at which of the 7 prompts drove this gap.

9. How I Would Scale This Further

- More categories: extend beyond apparel/handicrafts to electronics, groceries, and services, each as its own leaderboard view.
- Bigger annotator pool: route rating tasks through Josh Jobs — the same platform Josh Talks already uses for transcription QC — to reach hundreds of raters instead of 8–10.
- Automated model pulls: connect model APIs directly so new checkpoints are automatically queued for evaluation the day they ship, rather than generated manually.
- Admin audit trail: log every configuration change and rating edit — necessary once this feeds real model-selection decisions rather than a one-off exercise.

The current architecture (blind rating tool + live dashboard + leaderboard, all backed by a simple shared data store) already supports this scaling without a rebuild — adding a new model or category is a configuration change, not new engineering.